Exam 2
CpE 3150-Intro to Microcontrollers and Embedded Systems

This exam is an individual activity. The exam is open textbook and other class materials posted on Canvas for this course. Other outside materials are NOT permitted. Answers given that are identified as coming from outside sources or as shared work with another classmate will treated as academic dishonesty. All problems with answers identified as dishonest will be scored as 0 points, and there may be a referral to the ECE Department Chair for academic dishonesty.

Show all work. Partial credit will be given based on work shown.

I completed this exam on my own following the guidelines above and did not obtain or provide assistance to anyone else taking this exam. This includes not seeking online assistance from others.

_____ _____

Signature Date

(70 points) Part 1 of the Exam is a quiz (Exam 2: Part 1) on Canvas. This part of the exam consists of True/False and Multiple Choice problems. No work is required to be shown for this part of the exam. You may submit any work for this part of the exam on separate sheet(s) with the problem(s) labeled.

(30 points) Part 2 of the Exam is given as follows. Showing your work is necessary for partial credit. You may print out this part of the exam and write your solutions, scan them, and submit the scanned document on Canvas for Exam P. You may also provide your solutions (with work shown) in this Word document and submit on Canvas.

Both parts of the exam are subject to the integrity statement that you are signing above. Both parts of the exam must be completed (part 1 on Canvas) and submitted on Canvas (part 2 here) by the exam deadline. Best wishes with the exam!

1.Assume that PORTC is an input port connected to a temperature sensor. Write an assembly language program to read the temperature and compare it to the thermostat value of 65 (decimal). If the temperature value is above 65, set PD0 to activate a solenoid to turn on the air conditioner. Otherwise, clear PD0 to deactivate the solenoid to turn off the air conditioner. Write code to initialize PD0 as set (1) or cleared (0).

```
PC input
PD0 output
65 threshold
.EQU VALUE=66 ; USE 66 for same or high comparison
.ORG 0
LDI R16, 0x00
OUT DDRC, R16
LDI R17, 0xFF
OUT PORTC, R17; PC input
SBI DDRD, 0 ; PD0 output

Main: IN R18, PINC
      CPI R18, VALUE;
      BRSH TURNON ; current temp is > 65
      CBI PORTD, 0; temp ≤ 65 turn off A/C
      RJMP Nxt
TURNON: SBI PORTD, 0; turn on A/C
Nxt:  RJMP Main
```

2. **(10 Points)** Write an assembly program using Timer 0 to continuously generate a 18 kHz square wave on PORTD, bit 1 (PD1) using the CTC mode for Timer 0 with fosc = 8 MHz. SHOW YOUR WORK FOR TIMER PERIOD CALCULATIONS FOR FULL CREDIT.

$f = 18\,kHz$

$T = \frac{1}{f} = \frac{1}{18(10^3)}\,s = 0.0000555\,s = 55.5\,\mu s \Rightarrow 56\,\mu s$

$T_{half} = 28\,\mu s = t_{delay}$

$t_{delay} = n\,T_{MC}$ $\quad T_{MC} = \left(\frac{1\,cc}{1\,MC}\right)\left(\frac{1\,s}{8(10^6)\,cc}\right) = 0.125\,\mu s/MC$

$n = \frac{28\,\mu s}{0.125\,\mu s/MC} = 224\,MC < 256$ so no clk prescaler needed

PD1 output

```
.ORG 0
LDI R16, HIGH(RAMEND)
OUT SPH, R16
LDI R16, LOW(RAMEND)  move SP since Using subroutine
OUT SPL, R16
SBI DDRD, 1 ; PD1 output
LDI R16, 223 ; 224-1 For CTC
OUT OCR0, R16
LDI R17, 0
OUT TCNT0, R17 ; TCNT0 = 0 For CTC
LDI R16, 0b0000 1001 ; CTC, clk
OUT TCCR0, R16 ; turn on Timer 0, CTC, clk
LDI R18, 0b0000 0010 ; mask For bit 1
Here: IN R19, PORTD
      EOR R19, R18
      OUT PORTD, R19 ; toggle PD1
      RCALL Delay
      RJMP Here
.ORG 0x50
Delay: IN R20, TIFR
       SBRS R20, OCF0
       RJMP Delay
       LDI R21, (1<<OCF0)
       OUT TIFR, R21
```

3) **(10 Points)** Write an assembly language program to continuously read the value of PC7 and increment the count of 1s detected to PORTA. When the count in PORTA is greater than or equal to 75, reset PORTA to 0. Note that PC7 must be configured as an input and PORTA must be configured as an output port. DO NOT USE TIMERS FOR THIS PROBLEM.

PC7 input
PA output

75 count threshold
reset to 0

```
.EQU THRESH 75

.ORG 0
CBI DDRC,7
SBI PORTC,7 ; PC7 input + pullup
LDI R16, 0xFF
OUT DDRA, R16 ; PA output
LDI R18, 0 ; initialize count for A
OUT PORTA, R18
HERE: IN R19, PINC
      SBRC R19,7 ; PC7=0? skip
      INC R18
      CPI R18, THRESH ; count ≥ 75?
      BRSH resetCount
      RJMP OUTPUTCOUNT
resetCount: LDI R18,0 ; reset count
OUTPUTCount: OUT PORTA, R18; output count to PA
      RJMP HERE
```